

It is not immediately self-evident how you get from $|\mathbf{A}||\mathbf{B}|\cos\theta$ to $A_xB_x + A_yB_y + A_zB_z$. Although I can prove it when I want to,³ it takes me too long, so I remember them both.

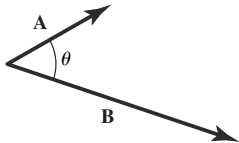


FIGURE 1-10 The vector dot product $|\mathbf{A}||\mathbf{B}|\cos\theta$ is the same in all coordinate